

Overconfidence of Q-learning agents in noisy signals in a voting environment

Noor Goudrar

Applied Economic Analysis Data Project

ECOI31237

April 15, 2026

1 Introduction

Research on the quality of information received by voters and its social consequences has been very thorough. Theories such as the Condorcet Jury Theorem state that a majority of a group is more likely than a single individual to choose the better of two alternatives, while the work of Toka (2008) emphasizes that a better informed electorate helps produce greater collective welfare. Thus, the collective ability to identify optimal policies is linked to the quality of information aggregation.

However, the emergence of online information, through online press and social media, has given rise to new challenges. The production of information has been significantly damaged as content can be relayed among users with no third party filtering or fact-checking. An individual user with no track record can in some cases reach as many readers as established and notable media. This shift led to an increasing political polarization, noted by Allcott & Gentzkow (2017) where they show that the diffusion of fake news played a significant role in the 2016 U.S. presidential election.

AI agents are increasingly used to personalize media selection and optimize user engagement. In this setting, understanding the velocity of the propagation of fake news through algorithms becomes crucial.

This paper studies the behavior of a Q-learning agent when exposed to noisy signals, representing biased information. We analyze if the agent develops overconfidence in the perceived signal, or if the uncertainty leads to more cautious behavior.

This study builds on the works of Kartal and Tyran in their paper “Fake News, Voter Overconfidence, and the Quality of Democratic Choice” (2022). The paper shows that overconfidence of voters in the signals they receive acts as a multiplier of fake news. While misinformation alone is harmful, its negative impact is exacerbated when voters are overconfident in their ability to distinguish truth from bogus information. Building on the Dunning-Kruger effect, the authors show that overconfident individuals fail to recognize their lack of competence. Consequently, they are more likely to vote based on misinformation rather than abstain. The paper also demonstrates that third parties can successfully share sponsored news to guide public opinion, if and only if the electorate is sufficiently overconfident.

In this assignment, we extend the setting of Kartal & Tyran to Q-learning agents. The goal is to understand how the agents’ confidence in the perceived signal evolves through reinforcement learning and if it can reach neutrality, if so under which conditions.

We show that neutrality of the agent is achieved only in the setting of perfect feedback on the quality of its perception and where overconfident-induced mistakes are penalized.

2 Model

This framework corresponds exactly to the model introduced by Kartal and Tyran (2022). A Q-learning agent must choose one of two policies: a and b . "Each agent votes for one of the two policies or abstains. The utility of the agent depends on the chosen policy and an unknown state of the world, $S \in \{A, B\}$. Thus, policy a is superior in state A and policy b is superior in state B .

Each agent i privately observes a noisy news signal, $s_i \in \{\alpha, \beta\}$, regarding the state. The term q_i denotes the precision of agent i 's signal; that is, $q_i = \Pr(s_i = \alpha | S = A) = \Pr(s_i = \beta | S = B)$. For every i , q_i is private information and an i.i.d. draw from distribution." The assumption in Kartal and Tyran is that "individuals differ in their competence and their ability to discern the true state, especially because they differ in their cognitive (e.g., critical thinking and analytical reasoning) abilities." In our setting, Q-learning agents do not have cognitive abilities. However, we keep the heterogeneity in signal precision as q_i is drawn randomly. We can thus still interpret q_i as a measure of signal precision.

"Agent i is either *biased* or *unbiased* in their perception of q_i . More specifically, i perceives q_i as $p_i(q_i)$, and $p_i(q_i) = q_i$ if i is unbiased, and $p_i(q_i) \neq q_i$ if i is biased. Biased agents are either overconfident or underconfident. If i is overconfident, then $p_i(q_i) > q_i$, whereas $p_i(q_i) < q_i$ if i is underconfident." In our model, we assume that the parameter λ acts as a multiplier on the perceived signal, such that $\pi(q_i) = \lambda \times q_i$. Thus, the agent's confidence level is defined by the value of λ :

- $\lambda > 1$ represents an *overconfident* agent;
- $\lambda = 1$ represents a *neutral* agent;
- $\lambda < 1$ represents an *underconfident* agent.

3 Q-learning

To replicate the environment stated above, the Q-learning model is very useful as for each experiment, the q-learner adapts its behavior to past experiences, taking actions to maximize its reward, allowing us to understand if the agent adopts an overconfident behavior or not. More precisely, we model two different Q-learning programs. In the first, the agent receives feedback on its signal precision after each episode. The second model is built on two Q-learning program, the agent only receives the noisy signal on the state of the world and must first form a belief on the signal precision. Here, we want to test if, while knowing its signal is noisy, an agent can still become overconfident. In the first model, an agent observes q_i and then chooses a confidence level $\lambda \in \{\lambda_O, \lambda_N, \lambda_U\}$ corresponding to overconfidence, neutrality and underconfidence. In the second model, the agent only observes s_i and chooses $\lambda \in \{\lambda_O, \lambda_N, \lambda_U\}$. In both models, the agent receives a reward, 1 if it voted for the correct policy and 0 if not. However, initial simulations showed that the Q-learner consistently chose overconfidence. This is explained by the absence

of a specific cost for overconfidence: within a maximization framework, the agent is not penalized for incorrect votes induced by overconfidence. To address this, we implemented a penalty specifically for cases where a overconfident-induced vote is verified. This is consistent with the conclusions of Kartal and Tyran where an overconfident agent who makes a mistake is responsible for a negative social cost.

4 Analysis

4.1 Behavior of the Q-learning agent

We present the results of four simulations with the variation of two parameters: whether signal quality q_i is revealed to the agent after each episode, and whether overconfident mistakes are penalized.

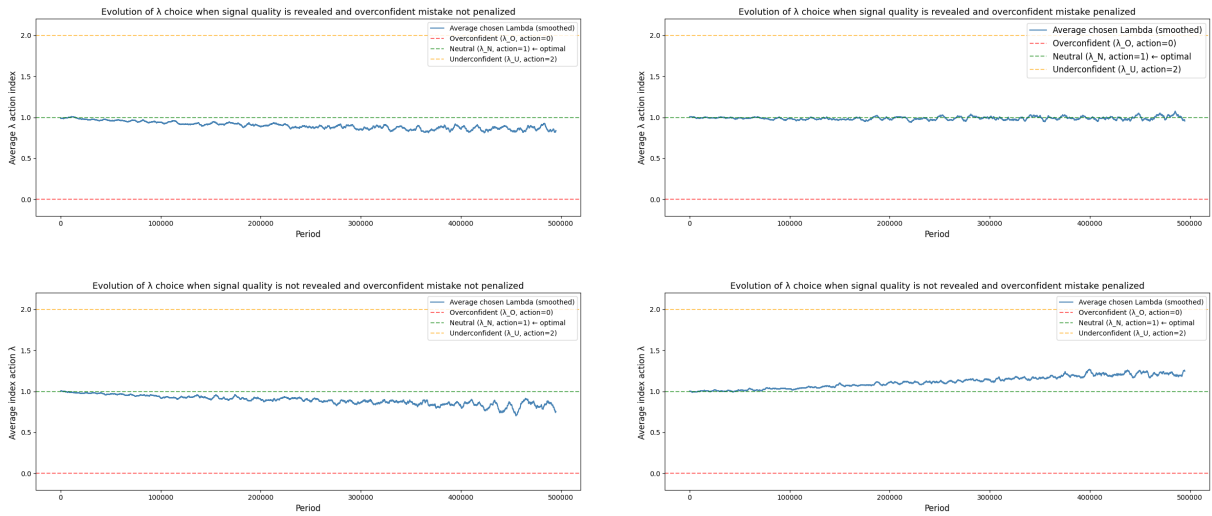


Figure 1: Changes in the confidence of Q-learning agents based on the type of feedback and penalty.

The figure shows the evolution of the action chosen by the agent throughout the experiment. In the two scenarios without penalty, we can see that the Q-learning agent converges to overconfidence, whether it has feedback on the quality of the signal or not. This stems from the absence of penalty due to overconfidence that encourages the Q-learning agent to maximize its reward. As overconfident mistakes are not penalized, the Q-learning agent cannot learn to avoid this action even if we show the quality of the signal each time.

However, we can see that in the two scenarios with penalties applied to overconfident mistakes, the Q-learning agent either learns to be neutral towards signal precision or even becomes underconfident.

Thus, we can say that the key determinant to achieve neutrality is penalizing the overconfident mistake. Even if the quality of the signal is showed to the Q-learning agent, it will still develop an overconfidence to the quality of the signal received.

4.2 Convergence of $p_i(q_i)$ to q_i

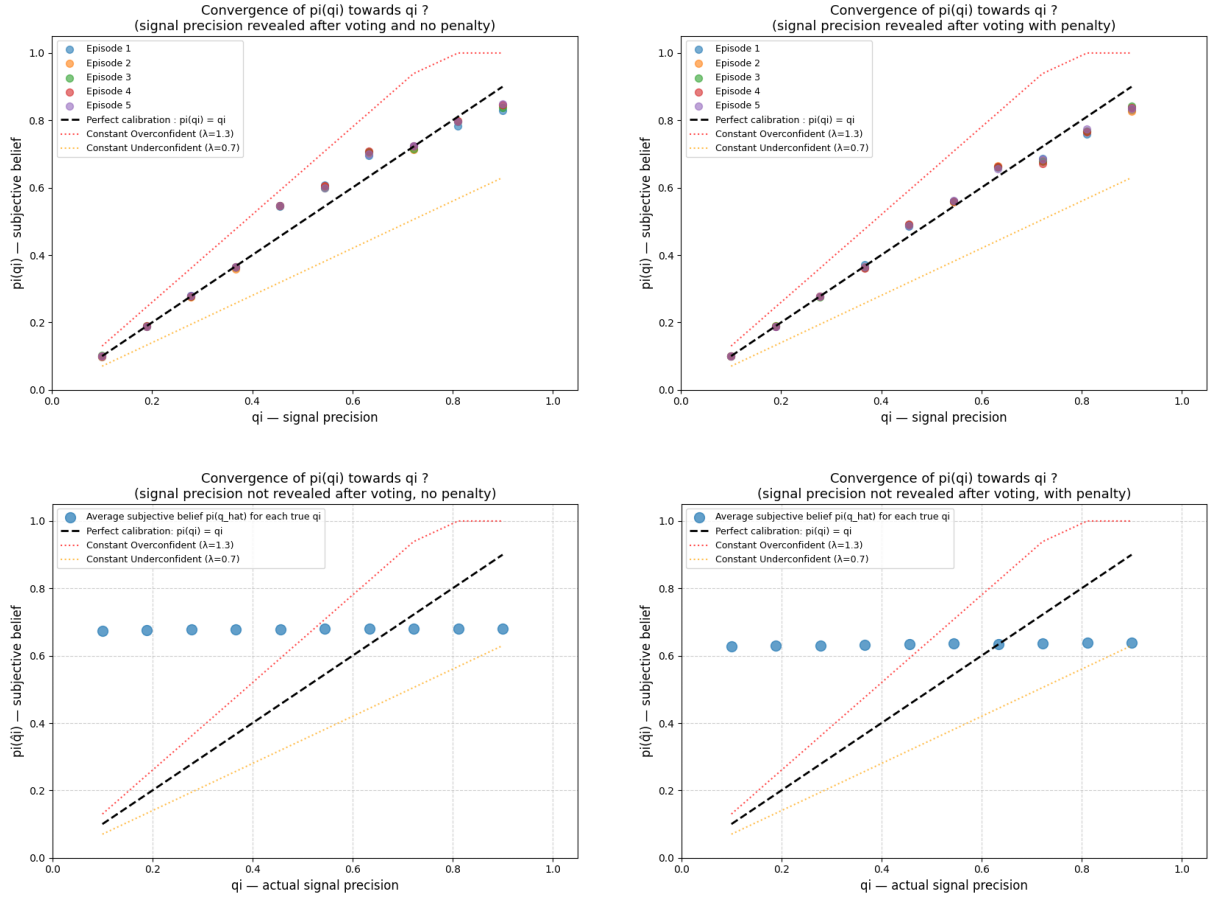


Figure 2: Convergence of subjective beliefs towards true signal precision.

In this figure, we want to understand if the subjective belief $p_i(q_i)$ ever converges to the true signal precision q_i . In the two cases where feedback is given on the signal, we plot signal precision on the x-axis and the subjective belief on the y-axis. In the cases where q_i remains hidden, the agent forms a belief $\hat{q}_i$ on the quality of its signal. To compare the agent's performance at calibrating its belief, we store the initial q_i drawn and compare the two.

The first striking difference here is that the Q-learning agent that receives no feedback on the quality of its signal does not change its subjective belief. Indeed, in both cases, with and without penalty, the subjective belief $p_i(\hat{q}_i)$ does not change relative to q_i . However, in the two other cases where signal precision is revealed, the Q-learning agent learned how to calibrate its subjective belief to match the true signal precision but still not perfectly.

4.3 Average reward over time

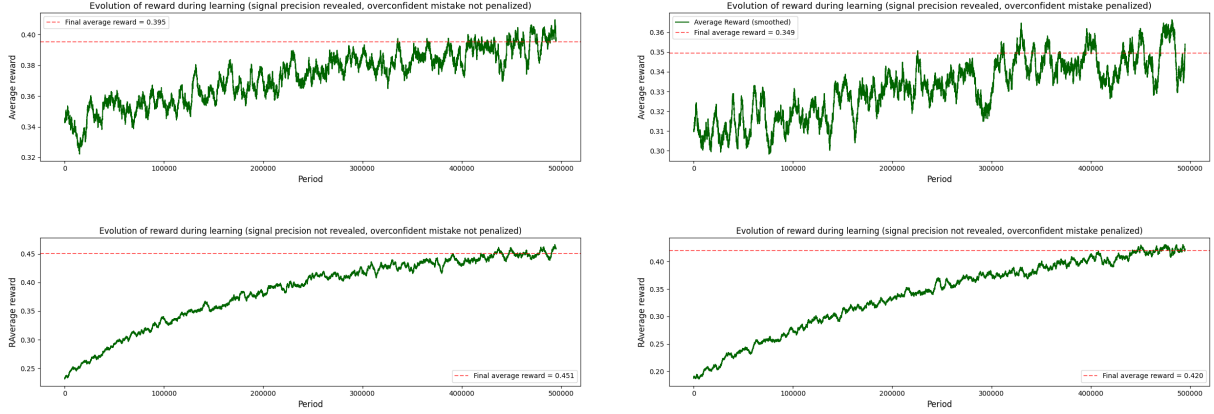


Figure 3: Evolution of reward during each period.

In this figure, we illustrate the reward evolution across periods. First, we can see that the reward is maximized for the cases where signal precision is not revealed to the agent. Second, the difference between the rewards when the overconfidence is punished or not can be explained by the penalty we applied. Still, it is interesting to understand why a blind agent would get better rewards. In the figure 2, we collect information on the subjective belief of the blind agent. The blind agent's subjective belief remains relatively constant regardless of the true signal precision, this allows it to focus its maximizing strategy on the reward.

We can explain the volatility of the reward in the revealed signal cases as the Q-learning agent adjusting its action right after receiving the information it was overconfident. This aligns with the conclusion of Figure 1, when signal quality is revealed and overconfident mistakes are penalized, the Q-learning agent converges to choosing the neutral belief. However, when signal precision is revealed and overconfident mistakes are not penalized, the Q-learning agent still adopts a more cautious behavior that does not maximize its reward, that is to say it is sensitive to noise. By design as there is no error penalty, it still chooses the overconfident action.

The key element here is that keeping the Q-learning agent blind to signal precision maximizes the reward. This setting is the closest to a real-life one where algorithms rarely get perfect feedback on the quality of their signal, and operate to maximize user engagement, associated here with the reward.

4.4 Change of behavior before and after Q-learning

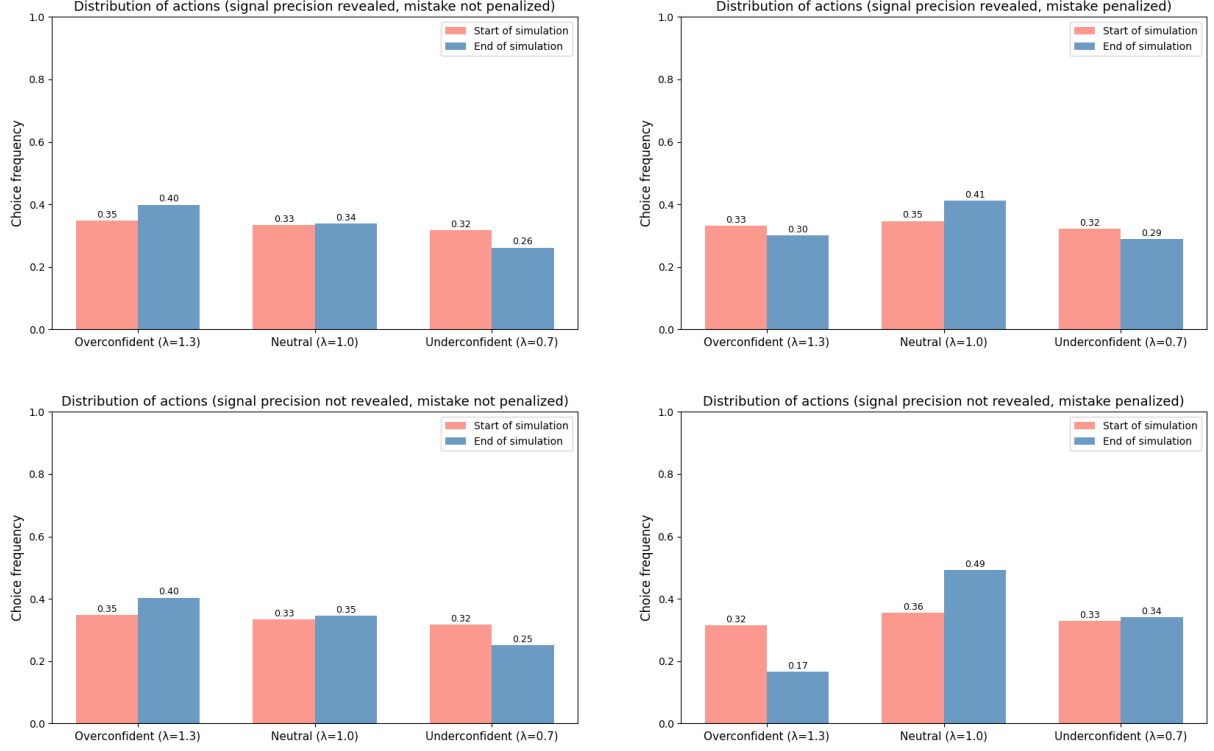


Figure 4: Evolution of actions before and after experiment

The figure illustrates the evolution of actions taken by the Q-learning agent before and after experiment. The cases where the agent learns to become overconfident are the ones where no penalty is applied to overconfident-induced errors. As a maximizing agent, the Q-learner cannot learn to calibrate its confidence in the signal as overconfidence is the most profitable strategy of the 3 actions. Although we have shown in Figure 3 that it is not the setting (among the 4) where the reward is maximized. When a social cost for overconfident-induced mistakes is applied, the agent shifts to a neutral belief in the signal precision, whether it was revealed to it or not.

5 Conclusion

Building on these four figures, we can conclude that the Q-learning agent only adopts a neutral confidence in the received signal when a penalty for overconfident-induced mistakes is applied. Otherwise, when no penalty is plugged in the model, the Q-learning agent becomes increasingly overconfident as this strategy does not affect its rewards.

As for the rewards, the settings where they are maximized are the ones where signal precision is not revealed to the Q-learning agent. Indeed, Figure 2 shows that blind agents keep the same subjective belief. This monotony allows the Q-learner to effectively maximize its rewards whereas when the signal precision is revealed, the Q-learning agent is exposed to more noise.

Hence, for a Q-learning agent to reach neutral perception of the signal it receives, a penalty for overconfidence must be added. It is the most effective feature to reach neutrality as we have shown that giving feedback on the signal precision doesn't necessarily make the Q-learning agent converge to a neutral stance.

These conclusions can help us design more truth-seeking algorithms operating specifically in social media and political information sharing. The spread of misinformation can be associated to our case where the signal precision is not revealed and where no penalty to overconfident-induced mistakes is applied. We have shown that this case generates the most overconfidence. To limit the spread of misinformation, Allcott and Gentzkow (2017) argue that addressing the social costs of fake news requires both increasing the information available to human agents about signal quality and strengthening the incentives to seek truth.

Looking back on the conclusions of Kartal and Tyran (2022), they state that voters' overconfidence in the signals they receive acts as a multiplier of fake news. Our model extends their experiment and allows us to see that the setting where the agent converges the most to overconfidence, is the one where signal quality is not revealed and no penalty is applied. Thus, if we aim to build neutral news algorithms, they should be calibrated to maximize veracity. For example, this could be achieved through cross-checking signal precisions before choosing to share the information or not, with a necessary penalty applied to the cases where the Q-learning agent shared a fake news with overconfidence.

References

- [1] Allcott, H., & Gentzkow, M. (2017). Social media and fake news in the 2016 election.
- [2] Kartal, M., & Tyran, J.-R. (2022). Fake news, voter overconfidence, and the quality of democratic choice.
- [3] Toka, G. (2008). Citizen information, election outcomes and good governance.